Unit: 2 [3 hrs]
Sound/Audio system

2.1 Concept of sound system

2.2 Music and speech

2.3 Speech Generation

2.4 Speech Analysis

2.5 Speech Transmission

Old Questions

- Explain speech generation. How does sound be represented digitally? Explain
- Describes different attributes of sound wave. Also describe MIMD

Sound/Audio:

sound/audio:

Audio refers to the audio content within a multimedia. Audio might be music, narration, sound effects, dialogues, ambient noise, etc. The use of audio in multimedia is to enhances the user experience by adding emotion, clarity to visual content.

Example: The background music in a video, voice-over in a tutorial, or sound effects in a game.

Audio are in two form:

- Analog audio:

A continuous signal that represents sound waves directly, mimicking their natural form. Analog audio are natural sound, that comes from real world. It is in continuous nature.

Example:

The output from a traditional microphone before it’s converted to digital.

- Digital audio:

A discrete signal that represents sound waves that is stored in the form of 0 and 1 is known as digital sound, Digital audio converts smooth sound into a set of numbers that computers can store and process, transmit easily.

Example:

A song downloaded from Spotify, a podcast file, or voice message.

File format of audio

Audio files come in different formats depending on their quality, compression, and use-case. Some are compressed to reduce file size, while others are uncompressed to maintain high quality. The various file format are as follows:

MP3 – MPEG-1 Audio Layer 3

WAV – Waveform Audio File Format

AAC – Advanced Audio Coding

OGG – Ogg Vorbis

FLAC – Free Lossless Audio Codec

AIFF – Audio Interchange File Format

WMA – Windows Media Audio

M4A – MPEG-4 Audio

MP3:

- The commonly used audio file format is MP3 and it is

lossy file format. The lossy type file refers it loosed the data after compressing the audio. Such cannot regenerated as original. MP3 file format is a compressed file.

- It is the most common audio file format, good quality with small size and used in music players and online streaming.

WAV:

It is uncompressed type audio file format. No compression applied on WAV file. WAV is a high quality audio. Such file format audio have large file size. It is used in professional audio editing and CDs

AAC:

AAC file format is compressed an lossy. This file format audio files are better than that of MP3 at the same bit rate. Such audio file format is found in iTunes and iPhones.

FLAC:

This audio file format is compressed but lossless in nature. It is used for high-quality audio with no loss. The file size is smaller than wav. It is used by audiophiles.

AIFF:

This file format is uncompressed. It is used by apple’s high quality audio and music production

Music

Music adds an emotional dimension to multimedia experiences, creating moods, building suspense, or enhancing the overall atmosphere. It involves melody, rhythm, harmony, and often instruments or singing. Normally music is instrumental that is no words or have lyrics which is words sing to a tune)

Speech

Speech, in the form of narration, voice overs, or spoken words, can be used to guide users, provide explanations, tell stories, or create interactive elements. It is form of verbal communication used to convey information, ideas, or feelings. It focuses on language and meaning, not musical structure. For effective convey of message, it uses intonation, stress, and pauses to express tone or emotion.

Speech Generation

Speech Generation in Multimedia (Text-to-Speech - TTS)

- Speech generation is a crucial part of multimedia systems, where written text is converted into spoken words using computer algorithms. This process, known as Text-to-Speech (TTS), helps multimedia applications communicate more effectively by adding audio output alongside visuals or text.

How Text-to-Speech Works:

- Text Analysis
- The system first analyzes the input text to understand its structure and meaning. It considers punctuation, grammar, abbreviations, and other language rules to prepare the text for speech conversion.
- Phoneme Conversion
- The text is then broken down into phonemes — the smallest units of sound in a language (like "k", "a", "t" for "cat").

- Pronunciation and Intonation
- Rules are applied to decide how each phoneme should sound, including stress, pitch, and rhythm. This step makes the voice sound natural and expressive, not robotic.

- Waveform Generation
- A digital speech waveform is created based on the phonemes and pronunciation rules. This waveform represents the actual sound of the voice.

- Audio Output
- Finally, the waveform is turned into an audio signal (like an MP3 or WAV file), which can be played back through speakers or headphones.

Speech Analysis

- Speech analysis in multimedia involves extracting meaning and insights from audio recordings and spoken language within multimedia contexts, such as lectures, meetings, and video calls. This analysis can be used for various purposes, including sentiment analysis, understanding customer satisfaction, and even detecting emotions in speech.

Methods to analysis the speech

- Speech Recognition (ASR):
- This technology converts spoken language into text, forming the basis for further analysis.

- Natural Language Processing (NLP):
- NLP techniques are used to analyze the meaning, sentiment, and relationships within the text generated from speech.

- Sentiment Analysis:
- Determines the emotional tone or sentiment expressed in the speech (e.g., positive, negative, neutral).

- Keyword Spotting:
- Identifies specific words or phrases within the audio, often used in customer service to track key issues or trends.

- Voice Biometrics:
- Used for speaker identification and verification, ensuring the authenticity of voices in multimedia content.

- Speech Synthesis (TTS):
- Converts text into spoken language, enabling multimedia systems to communicate with users.

Speech Transmission

Speech transmission refers to the process of sending spoken audio from one point to another over a communication system, often in real-time. In multimedia systems it ensures that speech reaches the receiver clearly and quickly.

How Speech Transmission Works:

- Speech Input
- A user speaks into a microphone, and the analog sound waves are captured.

- Analog-to-Digital Conversion (ADC)
- The analog audio signal is converted into digital form so that it can be transmitted over digital networks.

- Compression & Encoding
- To reduce the file size and bandwidth usage, the speech is compressed using codecs like MP3, AAC, .

- Transmission
- The encoded data is sent across a network (e.g., the internet, Wi-Fi, or cellular) using communication protocols

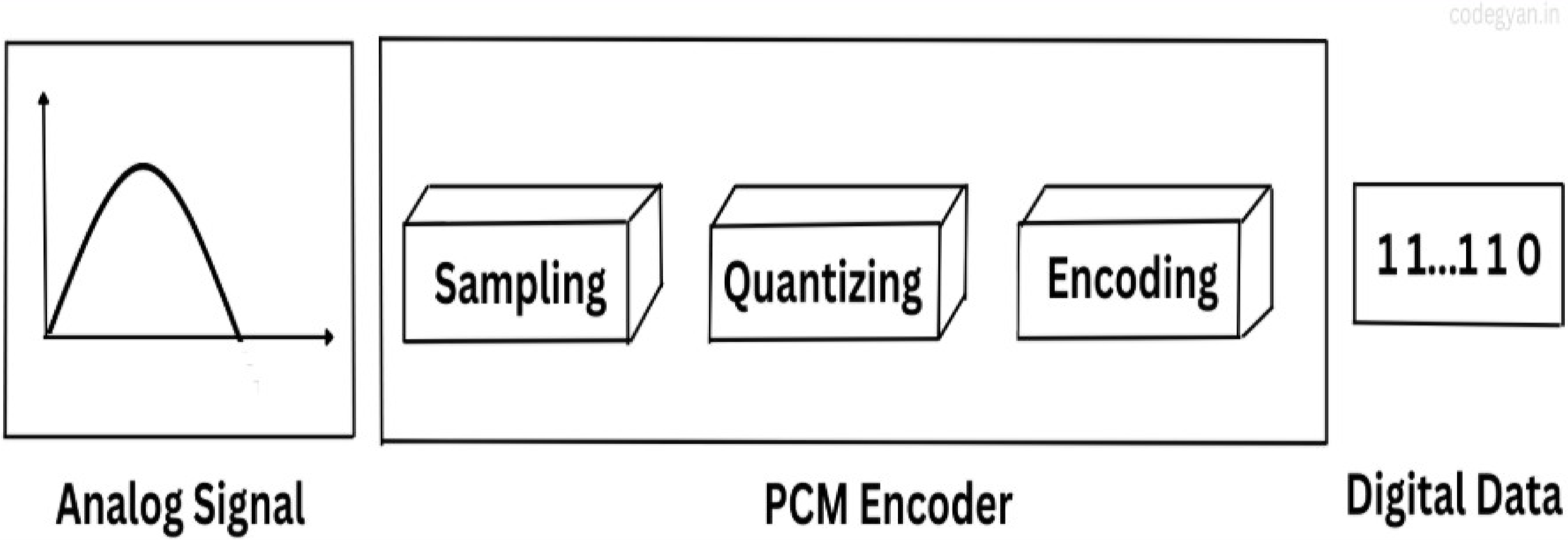
- Decoding & Playback
- At the receiver's end, the data is decoded, converted back into audio, and played through a speaker or headphones.

Digitization of sound

Digitalization of sound refers to the process of converting analog sound waves into digital signals that can be stored, manipulated, and reproduced using digital technology. Analog sound waves are continuous, meaning they are a continuous stream of electrical signals that fluctuate in amplitude and frequency. In contrast, digital signals are discrete, meaning they are a series of binary digits (0's and 1's) that represent the amplitude and frequency of the original sound wave.

• How Does Digitalization of Sound Work?

The process of digitalizing sound involves three main steps: sampling, quantization, and encoding.



1. Sampling: The first step in digitalizing sound is to sample the analog signal at a regular interval. This is done using an analog-to-digital converter (ADC), which measures the amplitude of the signal at regular intervals and converts it into a digital value. The rate at which the signal is sampled is known as the sampling rate, and it is typically measured in Hertz (Hz). The higher the sampling rate, the more accurately the digital signal will represent the original analog signal.

2. Quantization: The second step in digitalizing sound is to quantize the digital signal. This involves converting the continuous range of amplitude values into a finite number of discrete levels. The number of levels is determined by the bit depth of the digital signal, which is typically 16 bits or 24 bits. The more bits used, the more accurately the digital signal will represent the original analog signal.

3. Encoding: The final step in digitalizing sound is to encode the digital signal into a format that can be stored and processed by digital devices. This is typically done using a codec (coder-decoder), which compresses the digital signal into a smaller file size for storage and transmission.

Parameters of Sound

- Sound can be described using several key parameters: frequency (pitch), amplitude (loudness), and timbre (quality). These parameters are fundamental to how we perceive and understand sound.
- Elaboration:
- Frequency (Pitch):
- Determines how high or low a sound is perceived. It's measured in Hertz (Hz), which represents the number of sound wave cycles per second.
- Amplitude (Loudness):
- Refers to the intensity or volume of a sound. It's related to the pressure fluctuations in the medium through which the sound wave travels and is typically measured in decibels (dB).
- Timbre (Quality):
- Distinguishes the sound of different instruments or voices. It's the result of the complex combination of frequencies present in a sound and the way they interact.
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